MAHADEER A

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Research Profile

Doctoral researcher in Mathematics with research experience in Petri net theory, formal language theory, image segmentation, and image encryption. Seeking a postdoctoral fellowship to contribute to theoretical, computational, and interdisciplinary research while extending my expertise to related and emerging areas of mathematics and its applications.

Research Interests

Core Research Areas: Petri net theory, formal language theory, image processing, and information security.
Broader Research Interests: Discrete mathematical modelling, computational mathematics, and interdisciplinary applications of mathematics.

Education

- **Ph.D. in Mathematics (Thesis Submitted)** *Dec. 2022–Present*
  SRM Institute of Science and Technology Kattankulathur, Tamil Nadu
  - **Thesis:** *A Study on Petri Net Models for Partial Languages, Image Segmentation, and Image Encryption*
- **M.Sc. Mathematics** *2022*
  The New College, University of Madras Chennai, Tamil Nadu
  - **CGPA:** 9.084; First Class with Exemplary; **Gold Medalist**
- **B.Sc. Mathematics** *2020*
  The New College, University of Madras Chennai, Tamil Nadu
  - **CGPA:** 8.566; First Class with Distinction
- **Higher Secondary Certificate (HSC)** *2017*
  MEASI Matriculation Higher Secondary School Chennai, Tamil Nadu
  - **Percentage:** 83.16%
- **Secondary School Leaving Certificate (SSLC)** *2015*
  MEASI Matriculation Higher Secondary School Chennai, Tamil Nadu
  - **Percentage:** 93.20%

Research Experience

- **Doctoral Researcher** *Dec. 2022–Present*
  Department of Mathematics, SRM Institute of Science and Technology Kattankulathur, Tamil Nadu
  - Conducted theoretical research on Petri nets for generating partial and Watson–Crick partial languages, with emphasis on language classes and closure properties.
  - Developed Petri net-based approaches for image segmentation and image encryption, with the encryption frameworks incorporating cryptographic hashing and chaotic maps.
  - Published research articles in high-impact, peer-reviewed journals.
  - Collaborated with researchers on interdisciplinary studies leading to peer-reviewed publications.

Research Publication Summary

Total Research Publications:	08
Published Journal Articles:	07
Book Chapters:	01
International Journals:	*International Journal of Information Technology* (2); *IET Image Processing* (1); *New Mathematics and Natural Computation* (1); *Information Sciences* (1); *Nonlinear Dynamics* (1); and *Journal of Cybersecurity* (1).
Cumulative Impact Factor:	19.6
H-index:	2

Publications

S.No.	Publications	Indexing & Metrics
1	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "Enhancing Grayscale Image Encryption through Petri Net Dynamics and Hash-Extracted Keystreams," *Journal of Cybersecurity*, **12**(1) (2026), 1–19. https://doi.org/10.1093/cybsec/tyag019.	ESCI, Scopus, IF: 4.4, Q1
2	**Mahadeer A** and Arulprakasam R. "Petri Net-Driven Expanding Tent–Sine Chaotic Keystream Generator with Adaptive Token Flow for Image Encryption," *Information Sciences*, **748** (2026), 123515. https://doi.org/10.1016/j.ins.2026.123515.	SCIE, Scopus, IF: 6.0, Q1
3	**Mahadeer A** and Arulprakasam R. "Image Encryption Using Petri Net Dynamics with Pixel Permutation and Keystream Masking," *Nonlinear Dynamics*, **114**(8) (2026), 547. https://doi.org/10.1007/s11071-026-12413-y.	SCIE, Scopus, IF: 5.7, Q1
4	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "A Petri Net Token-Flow PRNG for Grayscale Image Encryption," *New Mathematics and Natural Computation*, 2026. https://doi.org/10.1142/S179300572850041X.	ESCI, Scopus, IF: 1.1, Q3
5	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "Adaptive Petri Net Token Flow Approach for Enhanced Grayscale Image Segmentation," *IET Image Processing*, **19**(1) (2025), e70255. https://doi.org/10.1049/ipr2.70255.	SCIE, Scopus, IF: 2.4, Q2
6	**Mahadeer A**, Arulprakasam R, and Dare V R. "Watson–Crick Partial Petri Net Languages and their Properties," *International Journal of Information Technology*, **17**(4) (2025), 2553–2569. https://doi.org/10.1007/s41870-024-02272-4.	Scopus, SNIP: 1.637, Q1
7	**Mahadeer A**, Arulprakasam R, and Dare V R. "Partial Petri Net Languages and their Properties," *International Journal of Information Technology*, **16**(6) (2024), 3663–3676. https://doi.org/10.1007/s41870-024-01903-0.	Scopus, SNIP: 1.637, Q1

Book Chapter

S.No.	Book Chapter	Publisher & Year
1	**Mahadeer A** and Arulprakasam R. "A Partial Petri Net Framework for Addressing Uncertainty in Healthcare Processes," in *Applied Mathematics, Automation and Computing*. https://doi.org/10.1201/9781003770497.	CRC Press, Taylor & Francis, 2026

Patent

S.No.	Patent	Application Details
1	**Mahadeer A**, Arulprakasam R, and Dhanalakshmi S. "Petri Net-Based Image Encryption Using Pixel Permutation and Keystream Masking."	Indian Patent Application No. 202641016722 A, 2026

Peer-Reviewer

Serve as an ad hoc reviewer for the following international peer-reviewed journals:

S.No.	Journal	Publisher
1	*Information Sciences*	Elsevier
2	*Information Processing & Management*	Elsevier
3	*Chaos, Solitons & Fractals*	Elsevier
4	*Journal of Information Security and Applications*	Elsevier
5	*Scientific Reports*	Nature Portfolio

Conferences

- Presented the paper entitled **"A Partial Petri Net Framework for Addressing Uncertainty in Healthcare Processes"** at the *CSIR & TNSCST-Sponsored International Conference on Applied Mathematics, Automation and Computing (ICAMAC 2025)*, organized by Kongu Engineering College, Perundurai, Tamil Nadu, India, February 21–22, 2025.
- Presented the paper entitled **"Modeling Uncertainty in Healthcare Process Management with Partial Petri Nets"** at the *12th National Conference on Mathematical Techniques and Applications (NCMTA-2025)*, organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, January 9–10, 2025.

- Presented the paper entitled **"Properties of Partial Petri Net Languages"** at the *Dr. Paarivendhar Research Colloquium (DPRC-2024)*, SRM Institute of Science and Technology, Kattankulathur, March 26–28, 2024.
- Presented the paper entitled **"Partial Petri Net Languages and their Properties"** at the *International Conference on Emerging Frontiers in Nonlinear Complex Systems, Computational Intelligence and their Applications (ICNCS 2024)*, organized by Vellore Institute of Technology, Chennai, February 7–9, 2024.

Workshops and Professional Development

- **Generative AI and Prompt Engineering** — Five-Day Virtual Faculty Development Programme organized by the Department of CSE (Emerging Technologies), SRM Institute of Science and Technology, Vadapalani, November 25–29, 2024.
- **Recent Frontiers in Pure & Applied Mathematics** — Five-Day Virtual Faculty Development Programme organized by the Department of Mathematics, Kongunadu Arts and Science College (Autonomous), under the DBT Star College Scheme, Coimbatore, November 11–15, 2024.
- **Research Proposal Writing for Funding Projects: Opportunities and Future Aspects** — International Workshop organized by Inspire Softech Solutions, Chennai, September 20–22, 2024.
- **Mathematics Behind Artificial Intelligence & Machine Learning** — Three-Day Workshop organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, May 16–18, 2024.
- **Mathematical Modelling and Its Applications** — International Workshop organized by the Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, December 13–14, 2023.
- **Advances in Graph Theory** — Five-Day Online Faculty Development Programme organized by the Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, September 11–15, 2023.
- **Role of Mathematics in Machine Learning** — Five-Day Faculty Development Programme for Skill Development organized by the Department of Mathematics, School of Basic Sciences, SRM Institute of Science and Technology, Kattankulathur, February 20–24, 2023.
- **Numerical and Analytical Techniques in Engineering Problems (IWNATEP-2023)** — Third International Workshop organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, January 30–February 1, 2023.

NPTEL Certifications

S.No.	Course Details	Certification & Score
1	**Introduction to Automata, Languages and Computation** IIT Kharagpur (12-week course; January–April 2023)	**Elite + Gold**; Score: 91%
2	**Graph Theory** IISER Pune (8-week course; January–March 2022)	**Elite + Silver**; Score: 77%
3	**Introduction to Abstract and Linear Algebra** IIT Kharagpur (8-week course; August–October 2021)	**Elite**; Score: 73%
4	**Introduction to Probability and Statistics** IIT Madras (4-week course; January–February 2021)	**Elite**; Score: 61%

Technical Skills

- **Document Preparation:** LaTeX (advanced proficiency)
- **Office Applications:** Microsoft Word, PowerPoint, and Excel (proficient)
- **Programming:** Python (working knowledge)

Honors and Awards

- **Dr. A.P.J. Abdul Kalam Research Award 2026** *February 2026*
  SRM Institute of Science and Technology
  - Conferred for excellence in doctoral research, based on a first-authored publication indexed in Web of Science/Scopus during the calender year 2025.

Additional Information

Gender: Male

Languages Known: English (speak, read, and write); Tamil (speak, read, and write); Arabic (read and write)

References

1. **Dr. R. Arulprakasam**
   Associate Professor
   Department of Mathematics
   College of Engineering and Technology
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2. **Dr. M. Rajesh Kannan**
   Associate Professor
   Department of Mathematics
   Indian Institute of Technology Hyderabad
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3. **Dr. Sairam Kaliraj**
   Assistant Professor
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   Indian Institute of Technology Ropar
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Declaration

I hereby declare that the information provided above is true and correct to the best of my knowledge.

MAHADEER A